

10MWRT ArUco Marker — Print Specification

Ground-reference marker for the 10-Metre Walk/Run Test (DM1 video assessment)

1. Marker details

Marker ID	0
Dictionary	ArUco 5×5
Printed size	175 × 175 mm (6.89 × 6.89 in) — print at 100% scale, no auto-fit
Paper	US Letter (8.5 × 11 in). A4 also works — do not let the printer auto-scale.
Detection library	OpenCV: <code>cv2.aruco.getPredefinedDictionary(cv2.aruco.DICT_5X5_50)</code> (or <code>DICT_5X5_100</code> / <code>_250</code> / <code>_1000</code> — all contain ID 0 with identical bit pattern)

Important: the detection script must use a **5×5** dictionary. If configured for 4×4, 6×6, or 7×7 the marker will not be recognised.

2. Print settings

- **Scale:** 100% / Actual Size. Disable “Fit to page”, “Shrink oversized pages”, and “Auto-rotate & center”.
- **Paper:** plain white matte printer paper, 80–120 gsm. Avoid glossy stock — it reflects ceiling lights and degrades contrast in video.
- **Colour mode:** black & white (grayscale). Pure black ink (K only) gives the best edge contrast.
- **Resolution:** any office laser/inkjet at ≥ 300 DPI is sufficient.
- **Orientation:** portrait. Do not rotate.

Always print from the original PDF. Do NOT photocopy a photocopy — generational loss rounds the square edges and degrades detection.

3. Verification after printing

- Measure the outer black square with a ruler. It should read **175 mm ± 2 mm** (6.89 in ± 0.08 in) on each side.
- The marker must be perfectly square — if width and height differ by more than 2 mm, the printer applied unequal scaling. Reprint with scaling disabled.
- All four corners of the black border must be sharp, fully inked, and unbroken. Any gaps in the border prevent detection.
- Quick software check: photograph the print straight-on and confirm it is decoded as **ArUco 5×5, ID 0** before use.

4. Lamination and mounting

- **No glossy lamination** — it causes camera glare. Use a *matte* pouch laminator only.
- **Rigid backing:** mount on rigid cardboard or 3 mm foamboard so the marker lies perfectly flat on the floor.
- **Do not cut inside the border.** The full black border must remain intact on all four sides.